

From a Physical Target to a Checkable Root

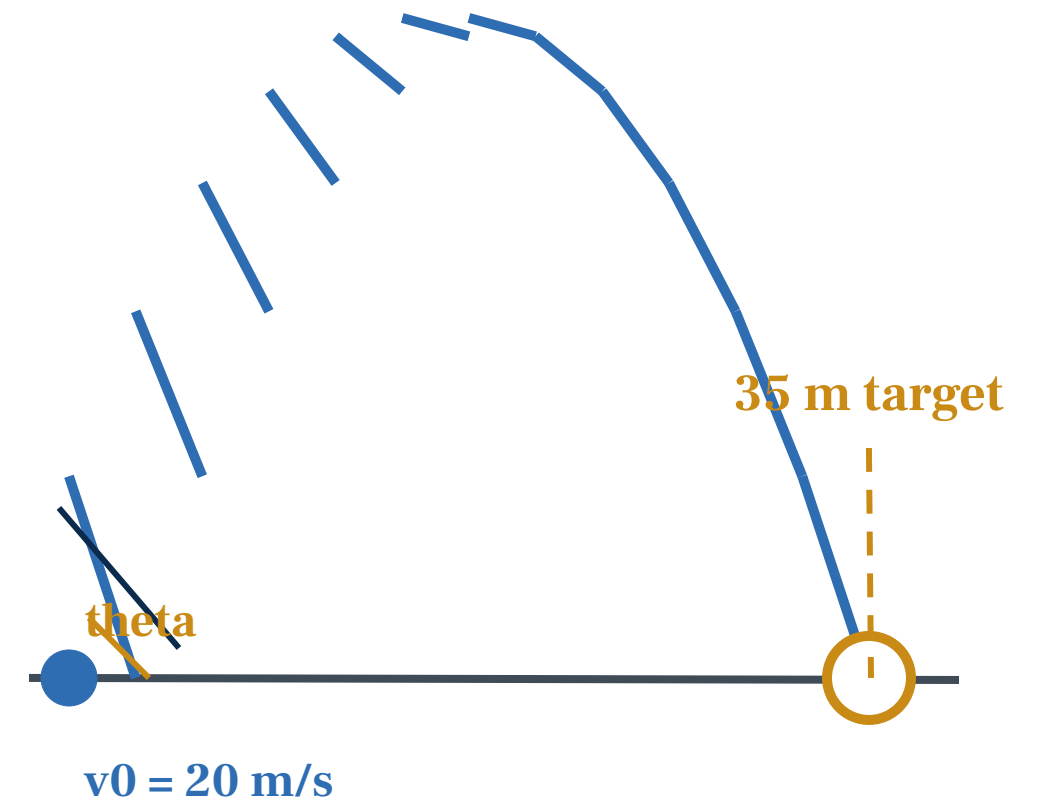
Week 5 | Root finding

Which launch angle makes the range equal to 35 m?

$v_0 = 20 \text{ m/s}$

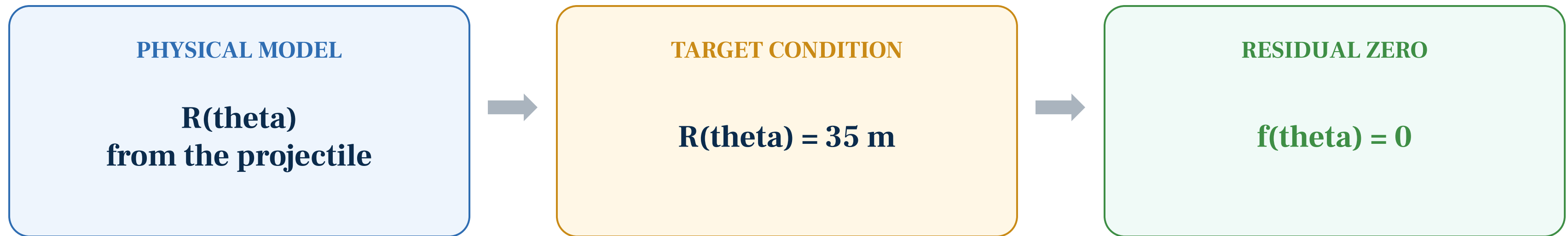
target range = 35 m

physical target $\rightarrow$ residual $\rightarrow$ root



A target becomes a residual

Model range minus target range becomes a quantity a solver can drive to zero



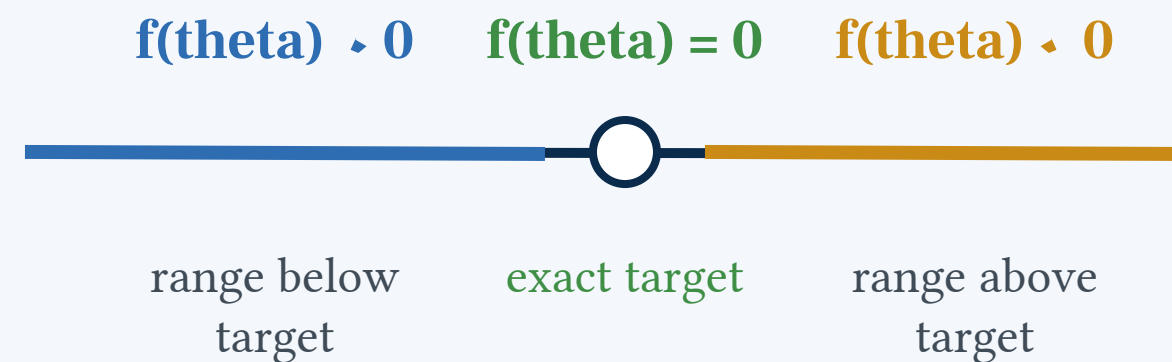
$$R(\theta) = \frac{v_0^2}{g} \sin(2\theta), \quad v_0 = 20 \text{ m s}^{-1}, \quad g = 9.81 \text{ m s}^{-2}$$
$$R(\theta) = 35 \text{ m} \quad \implies \quad f(\theta) = R(\theta) - 35 \text{ m} = 0$$

root variable: theta (degree) | residual: metres

The residual sign tells you which way to move

The sign is a physical error: below target or above target

Read the sign as a range error



On the low-angle branch

Turn sign into a direction



negative residual

the model falls short of 35 m



move to a larger angle on this branch



positive residual

the model overshoots 35 m



move to a smaller angle on this branch

The graph gives us a bracket

A sign change marks the low-angle interval we will keep

Scan before solving

$\theta = 10:1:70$ degree

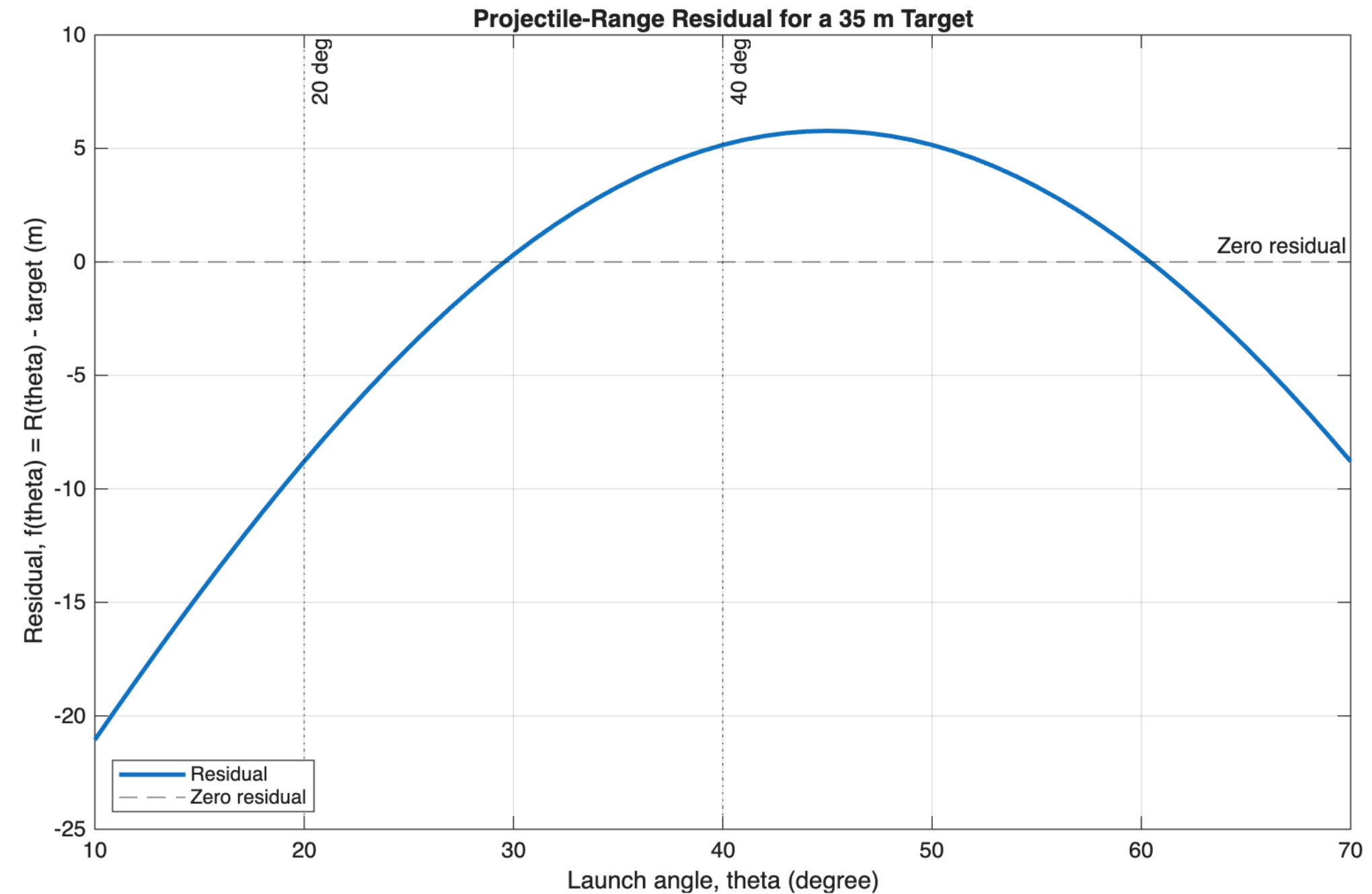
$f(20 \text{ degree}) = -8.7905 \text{ m}$

$f(40 \text{ degree}) = +5.1553 \text{ m}$

opposite signs

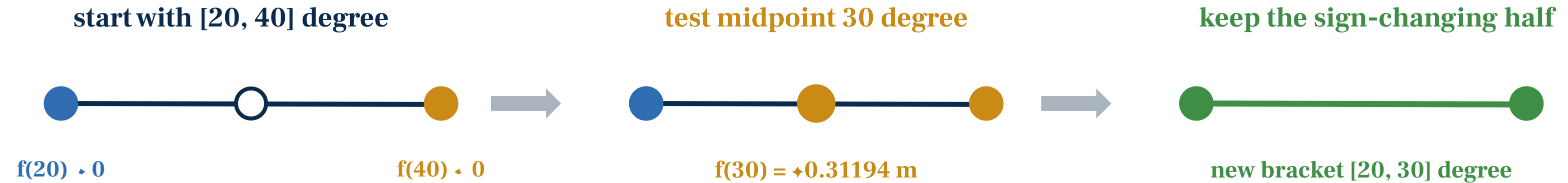
bracket = $[20, 40]$ degree

Where does $f(\theta)$ cross zero?



Bisection keeps a sign change alive

Bracket first, test the midpoint, keep the half that still straddles zero



- 1 choose endpoints**
opposite signs
- 2 compute midpoint**
 $m = (a + b)/2$
- 3 evaluate residual**
read $f(m)$
- 4 keep the sign change**
replace one endpoint

Why is the midpoint not enough?

Because bisection needs a bracket whose endpoint residuals have opposite signs.

Bisection shrinks the bracket

Every row repeats the same four decisions

Plain-language algorithm

- 1 Start with opposite endpoint signs
- 2 Compute the midpoint
- 3 Evaluate $f(\text{mid})$
- 4 Replace one endpoint
- 5 Repeat with the smaller bracket

invariant: $f(a)$ and $f(b)$ keep opposite signs

iteration	current bracket (degree)	midpoint (degree)	$f(\text{mid})$ (m)	keep
1	[20, 40]	30	+0.31194	[20, 30]
2	[20, 30]	25	-3.7648	[25, 30]
3	[25, 30]	27.5	-1.5993	[27.5, 30]
4	[27.5, 30]	28.75	-0.61095	[28.75, 30]

The bracket width halves; the sign change is the invariant

The scaffold makes each decision visible

The loop is short because the physical decision is explicit

Read the loop in plain language

- 1 choose left and right
- 2 compute midpoint
- 3 evaluate residual
- 4 if small, stop
- 5 otherwise keep the sign change

tolerance_m = 1e-3 m

MATLAB

```
mid_deg = (left_deg + right_deg)/2;
f_mid_m = (v0_mps^2/g_mps2)*sind(2*mid_deg) - target_m;
if abs(f_mid_m) < tolerance_m
    break
end
f_left_m = (v0_mps^2/g_mps2)*sind(2*left_deg) - target_m;
if f_left_m*f_mid_m < 0
    right_deg = mid_deg;
else
    left_deg = mid_deg;
end
```

Which line protects the bracket?

Stop once, then substitute back

A numerical angle matters only if it satisfies the original condition

One Core stopping rule

$\text{abs}(f_{\text{mid_m}}) \prec 1\text{e-3 m}$

The residual is an error in range, so the tolerance is measured in metres.

then validate the root

$$\theta_* = 29.5679 \text{ degree}$$

$$R(\theta_*) = 35.0004 \text{ m}$$

$$f(\theta_*) = R(\theta_*) - 35 \text{ m} = +4.1164 \times 10^{-4} \text{ m}$$

VALIDATION

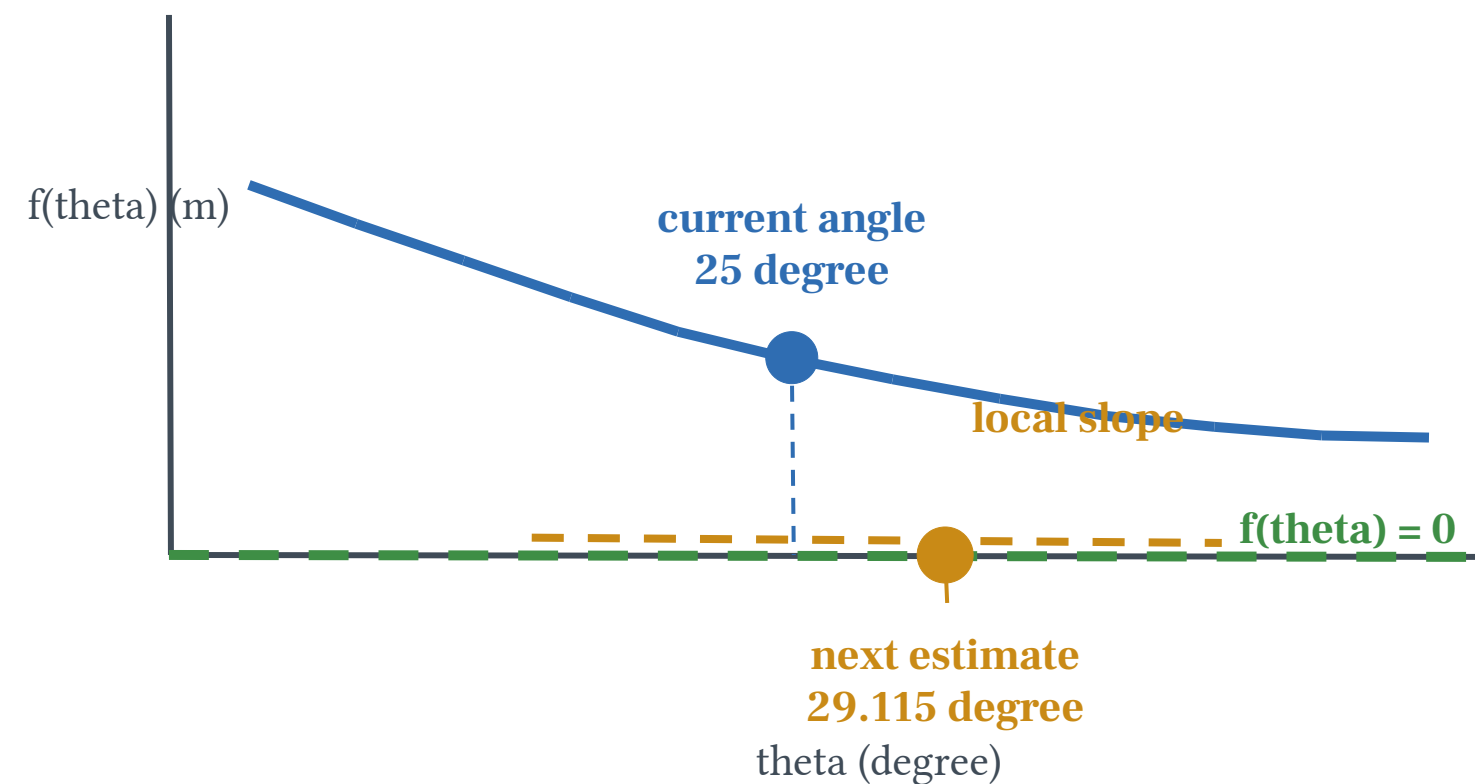
$\theta^* = 29.5679 \text{ degree} \mid R(\theta^*) = 35.0004 \text{ m} \mid f(\theta^*) = +4.1164\text{e-4 m}$

The root is useful because substitution returns the target range

Newton follows the local tangent

One estimate plus a local slope gives the next angle

A tangent predicts the next angle



$$\text{new angle} = \text{current angle} - \frac{\text{current residual}}{\text{current residual slope}}$$

$$\frac{df}{d\theta} = \frac{v_0^2}{g} \cos(2\theta) \frac{2\pi}{180} \quad (\text{metres per degree})$$

slope unit = metres per degree

What does the slope measure?

Newton can jump close to the root

Four evaluations move from 25 degree to a validated root

Trace the current estimate, residual, slope, and next estimate

eval	theta (degree)	f(theta) (m)	slope (m/degree)	next theta (degree)
1	25.000	-3.7648	0.91488	29.115
2	29.115	-0.33461	0.74939	29.562
3	29.562	-0.0042242	0.73044	29.567
4	29.567	-7.1308e-7	0.73019	stop

Each row

evaluate

↓

slope

↓

update

What changes between rows?

At eval 4, $|f(\theta)| = 7.1308e-7 \text{ m} \ll 1e-3 \text{ m}$

Working exposure: same root, different route

Bisection is guarded; Newton is locally fast

BISECTION

13

recorded
iterations

preserves a sign-changing bracket

needs a valid endpoint pair

NEWTON

4

recorded
evaluations

same low-angle root from 25 degree

needs a useful slope and start

Both still require residual validation

A midpoint is not enough: the bracket must change sign

The method only works when the endpoints still have opposite signs



CHECK AFTER EVERY UPDATE

$$f(a) * f(b) < 0$$

If the signs agree, the bracket is not safe

The residual pattern transfers to new physics

Change the model; keep the reasoning chain

physical target → residual → root → validation

THIN LENS

target: image distance

unknown: d_i

validate by substitution into the lens equation

RC THRESHOLD

target: threshold voltage

unknown: t

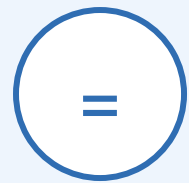
validate by checking $V(t)$ against the target

$$\text{thin lens: } f(v) = \frac{1}{f} - \frac{1}{u} - \frac{1}{v} = 0 \quad [f(v) \text{ in m}^{-1}, v \text{ in m}]$$

$$\text{RC charging: } f(t) = V_s (1 - e^{-t/(RC)}) - V_{\text{threshold}} = 0 \\ [f(t) \text{ in V}, t \text{ in s}]$$

Exit ticket: explain a root to a future you

Define it, bracket it, update it, check it



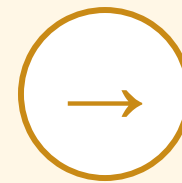
DEFINE

write the physical target
and residual



BRACKET

explain what opposite
endpoint signs mean



UPDATE

describe one bisection
midpoint
or Newton tangent step



CHECK

substitute the root into the
residual
and state its physical unit

practical transfer: projectile range • thin lens • RC threshold